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Convolutional Neural Network for Diabetic Retinopathy Detection

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***Abstract*—The world's projected blind population will reach 40 million by 2025. A modern fundus-based algorithm that approves the classification of retinal tissue needs to be improved in the early stages of healthy and diabetic retinopathy. In this experiment, we have introduced a convolution neural network approach to detecting diabetic retinopathy. We used the publicly accessible Apatos Blindness Detection database to train a convolution neural network, where the image is processed at an early stage, primarily involving image resizing, pixel rescaling, and label encoder. After that, an image is given to the convolution neural network model, to decide whether the patient has diabetic retinopathy or not. About 3789 color retinal images are used in experiments to train the proposed model and about 948 images are collected to test its efficiency in classification. Accuracy of 96.15\%, Sensitivity 79\%, Precision 89\%, and F1-Score 84.1\% and Area Under Score 0.82 is achieved using the Convolution Neural Network-based method.**

Keywords—convolutional neural network, diabetic retinopathy, deep Learning.

I. Introduction

Diabetes is the biggest issue being suffered by the world today. According to a World Health Organization survey, the global incidence of diabetes was 2.8\% in 2000 and by 2030 current statistics will rise by 4.4\% [1]. The direct adverse effect of a retina blood vessel (BV) is diabetic retinopathy (DR). Gloomy strings in sight, fuzzy sight, irregular sight, vitiated sight, vacant areas in sight, and difficulty seeing at night are perfect symptoms of DR [2].

Five types of DR are available: Mild, Moderate, Severe, Proliferative, and No DR. Mild nonproliferative Like a balloon, the small area of BV in the retina can swell. Moderate nonproliferative certain of the BV in the retina will be blocked. The severe nonproliferative growing number of BV nourishing the eye has become blocked. Proliferative is the ultimate stage of DR. This is when fresh BV begins to evolve in the retina. This is called neovascularization [3]. There are various lesions of DR: The most initial clinically noticeable DR changes are microaneurysms. Hemorrhages occur when weak capillary walls break down. Hard exudates are white yellow deposits ranging from small to large patches that evolve into circular rings. Greenish-white discoloration patches with indistinct (fluffy) edges are included in the soft exudate nerve fiber sheet [4]. (fluffy) edges are included in the soft exudate nerve fiber sheet [4].

Deep learning is an efficient instrument for predicting an actionable outcome. Convolution Neural Network (CNN) is mostly used for the grade of images [5]. CNN is commonly used by many researchers in medical image scrutiny than other DL-based methods. The architecture of CNN contains three popular layers: convolution, pooling, and fully connected layer. This paper is divided as follows: In section II, some literature survey is done. In section III, our approach, methodology, and flowcharts used for DR classification are shown. The results of the experiments are shown in section IV. Section V concluded the paper.

II. Literature Review

Many researchers have discussed the issue of highly accurate classification of images. Here are some citations relevant to our paper.

Kele Xu et al., [6] has developed an approach to DR detection using deep CNN. In this work, the authors used the publicly available data set Kaggle. Data augmentation Steps are used such as stretching, translation, flipping, And rotate to the labeled dataset. CNN's architecture is then used for automatic DR classification and achieves 94.5\% accuracy.

Vishakha Chandore et al., [7] have evolved a method for automatic disclosure of DR using deep CNN. A large dataset of around 35000 images is used. In this work, images are resized into dimensions of 448x448. They also applied various data augmentation steps. and Finally, the authors achieved a precision for class 0 is 81\% and for class 1 88\%.

Using the Kaggle dataset, Mohammadian et al., [8] proposed a framework for categorising DR into two classes: DR and No DR. The authors used a dataset of 35,126 images, of which 20\% images used to test the algorithm's output against unseen data. The authors tweaked the final two blocks of the two architectures and contrasted two optimizers with different LRs: stochastic gradient descent and Adam. To increase the system's accuracy, the authors augmented the images by flipping them horizontally and vertically or rotating and moving them. The accuracy of 87.12\% for the

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InceptionV3 architecture and 74.49\% for Xception is achieved.

Dinial Qomariah et al., [9] have suggested a method for the classification of DR and healthy retinal images using CNN and SVM. They used the publicly accessible Messidor database. Here features are obtained using transfer learning and accuracy of 95.83\% for base 12 and for inception v3 and VGG19 95.24\% for base 13 is achieved.

Harry Pratt et al., [10] have developed a model for diagnosis of DR using CNN. Here images are rescaled into 512x512 pixels. Here the Kaggle dataset is used. In this research, the CNN architecture can be built to define the dynamic characteristics involved. in the task of classification. They achieved a Recall of 95\% and 75\% of accuracy.

Nidhi Kamothi., [11] have suggested a method for Automatic Diagnosis of DR using Transfer Learning. Image preprocessing steps are used such as image enhancement and converted in the desired format. After that various CNN architectures are used like CNN, VGG16, and VGG19. They reported the accuracy of the model based on the VGG16 is 71.6\%.

III. PROPOSED METHODOLOGY

Different steps involved in methodology are given below:

- Get the Dataset.
- Data Preprocessing.
- The dataset is divided into two parts: training and testing.
- Data augmentation technique used.
- Use CNN algorithm for training the model.
- Get the evaluation matrices.
- Get predictions.
- Make a classification table.

A. Dataset

We have obtained the dataset from the Blindness Identification 2019 Asia Pacific Tele Ophthalmology Society (APTOS 2019 BD) [14]. APTOS has produced a comprehensive collection of retina photography fundus photographs that have been taken under a wide range of imaging conditions. The data was complex and comprehensive. The clinician graded each picture on a scale of 0 to 4 for the seriousness of diabetic retinopathy. We have split a dataset into two classes (class 0 and class 1). In 0 class NO DR images and in 1 class combined Mild, Moderate, Severe NPDR and PDR images as DR Images.

B. Preprocessing

As the dataset contains image data, it is first pre-processed by using various steps such as resizing, pixel rescaling, and label encoder.

1. Resizing: Images are resized to a pixel of 64x64. Fig. 1 shows Image Resizing.

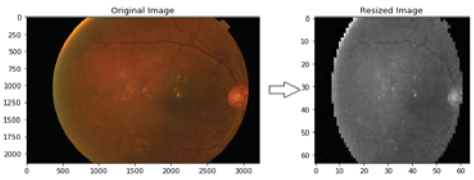

Fig 1. Image Resizing

2. Pixel Rescaling: Pixel values are rescaled into 0 and 1 for each image by dividing by 255 for ease of computation. It enhances the efficiency of the activation function. The Sigmoid function performs better with data between 0 and 1.
3. Label Encoder: The method of translating a label into a numeric format so that it can be translated to a machine-readable format is label encoding.

C. Dataset Splitting

Here the dataset is divided into two parts 80\% data used for training and 20\% data used for testing. The training set contains 3,789 samples, while the validation set contains 948 samples, 365 of which are stable and 583 of which are diabetic. We ensure that the images chosen for the test are not used during training in order to perform successfully the binary classification task.

D. Data Augmentation

It is used to resolve both the criteria, the variety of training data, and the volume of data. In comparison to these two, the augmented data is also used to resolve the problem of class disparity in the classification tasks [12]. After dataset preprocessing and splitting, a data augmentation strategy is used for the process of training. In addition, the methods we used include data cropping, data shifting, rotation, padding, flipping, etc. Data augmentation is shown in Table I.

TABLE I.

DATA AUGMENTATION USED

Argument	Parameter Value
Zoom Range	0.2
Width Shift Range	0.2
Rotation Range	50
Horizontal Flip	True
Fill Mode	Nearest

E. Classifier

Fig. 2. Represents essentially the complete architecture of the CNN model is simply divided into several layers such as convolution, pooling, and fully connected layers.

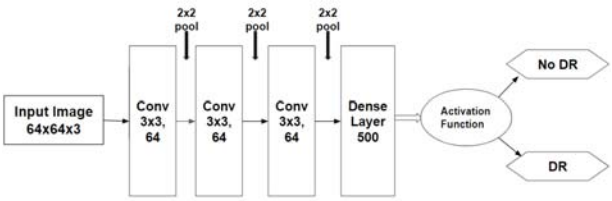

Fig. 2. CNN Architecture

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1. Convolutional Layer: It is the first layer, and this layer performs Heavy computing, which makes it easier for extra work. This layer functions as an input layer, with the image's input size set to 64x64x3.
2. Max pooling Layer: The highest weighted function is separated in the Max-pooling layer, which is accomplished by transforming the preceding 3x3 matrix into a more compact matrix. The above 3x3 matrix is then transformed into a 2x2 matrix that only includes the 3x3 matrix's highest weighted function.
3. Flatten Layer: The flattened layer reduces the image matrix to a single dimension, which serves as the dense layer inputs.
4. Dropout Layer: The dropout layer performs a higher and efficient operation that significantly enhances the network's normalization capabilities. This method uses random removal and potentially restores neurons during training with a probability determined by a hyperparameter called dropout rate.
5. Dense Layer: These layers serve as the final elements of a deep neural classifier, and they are fed by the attributes removed by the preceding convolutional layers.
6. Output Layer: Is the final layer of the network that generates the network's output, whether it's a Softmax layer or a sigmoid neuron, depending on whether the role is binary or multiclass classification.

Fig. 3. shows the Flowchart for designing an algorithm. After designing the algorithms, we have done classification based on four parameters: precision, accuracy, recall, and Fl-score.

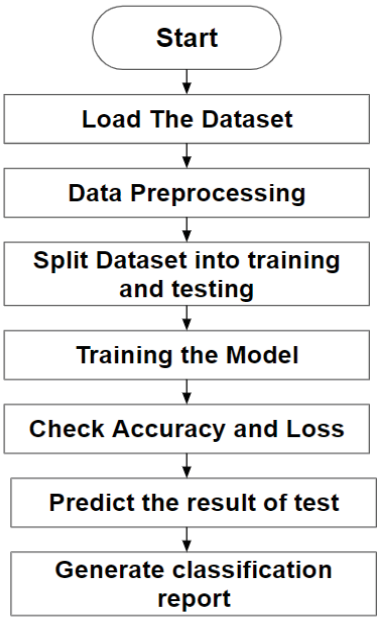

Fig. 3. Flowchart of Model

In the process of deep learning, optimizers play a key role. The optimizer for Root Mean Squared Propagation (RMS prop) is used here. The rate of learning is set automatically in RMSprop and a different rate of learning is chosen for each parameter. The learning speed was separated by RMS prop through the average square gradient exponential decay. The batch size here is set to 128 and the epoch is set to 50.

F. Performance Measures

1. Confusion Matrix: To get the idea about classification and misclassifications. The confusion matrix is the m x m matrix [15]. Let, m is the no. of variables involved. Diagonal elements represent the true values and false values are expressed by non-diagonal elements. There are four parameters inside the figure such as, False Positive (FP), True Negative (TN), True Positive (TP), and False Negative (FN). Fig. 4. demonstrates a confusion matrix.

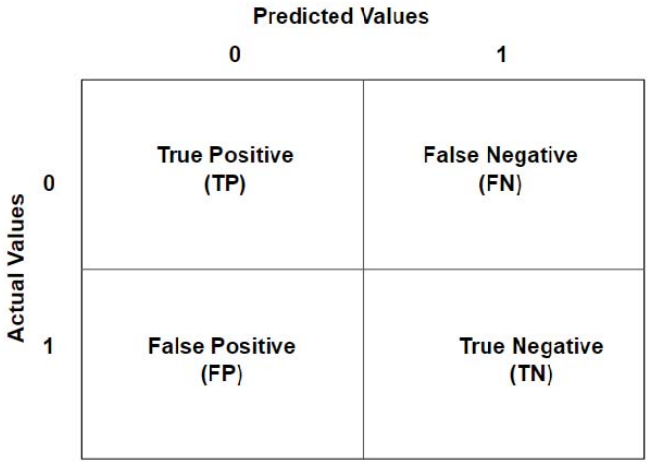

Fig. 4. Confusion Matrix

2. Accuracy: Accuracy is determined as a ratio of the events properly classified to the total events.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

3. Sensitivity or Recall: Sensitivity measures the degree of positively classified positive events.

$$Sensitivity = \frac{TP}{TP+FN} \quad (2)$$

4. Precision: Precision is determined as the ratio of the number of Positive samples correctly identified to the total number of Positive samples either correctly or incorrectly.

$$Precision = \frac{TP}{TP+FP} \quad (3)$$

5. F1-Score: F1-score is the arithmetic average of recall and accuracy.

$$F1\text{-}Score = 2*\frac{Precision * Recall}{Precision + Recall} \quad (4)$$

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IV. RESULTS

Results of the binary classification of CNN Architecture DR images provided in this section. Furthermore, the quality and functionality of the model should be checked. Its outcomes are presented in Fig. 6-7. Using validation and training curves of accuracy and loss. Other results are tabulated below Table II-IV.

The performance evaluation reports are obtained as shown in Table II.

TABLE II

PERFORMANCE EVALUATION REPORTS

Stages	Precision	Recall	F1-Score
Class 0	0.72	0.85	0.78
Class 1	0.89	0.79	0.84

Table III shows comparison results. Kele Xu et al.,[5] used 8 convolutions, 3 pooling, and 2 fully connected layers. 3 convolutions, 3 pools, and 2 fully connected layers were used in the proposed model, which would reduce the model's complexity. We trained the model on more datasets in this experiment to further increase the model's accuracy. We may adjust parameters such as epochs to optimize the output of the CNN model, the number of epochs decidingly affect efficiency. For a significant no. of epochs, there is an improvement in performance. Some experimentation is needed to determine the epochs. After a few epochs, there has been no decrease in training loss and an increase in training accuracy. Accordingly, a number of epochs can be determined.

TABLE III

COMPARISON OF RESULTS

	Kele Xu et al. [5]	Proposed Model
Size of training data	1000 Images	4,737 Images
Optimizer	Adam	RMSprop
Data Augmentation Used	Yes	Yes
Accuracy	94.5%	96.15%
Loss	Not Specified	0.0839

In Table IV Confusion matrix is shown the classification reports of the test datasets. It is observed that in the first image class (Normal), the model was able to identify 310 images correctly in the normal class, but 55 were labelled as DR. Likewise, for the second image class (DR), the model was able to identify 463 images correctly, but 120 images were labelled as Normal.

TABLE IV

RESULTS OF CONFUSION MATRIX

	Predicted Value	
	Class 0	**Class 1**
Actual Value	310	55
	120	463

Fig. 5. depicts the history of training and validation accuracy. Training accuracy from 80\% to about 94\% is improved up to 30 epochs, and after 30 epochs the accuracy is almost constant at around 96\%. After there are slight spikes in the graph from 75\% to 85\%, the validation accuracy improves significantly before 20 epochs. Accuracy is simply a metric that can only be applied to the task of classification. The exact classification of the proportion of your test data is defined. Model accuracy is normally chosen after examining and determining the model parameters and no learning takes place. The test samples are then given to the model, and the model's number of errors is reported after the actual targets are compared.

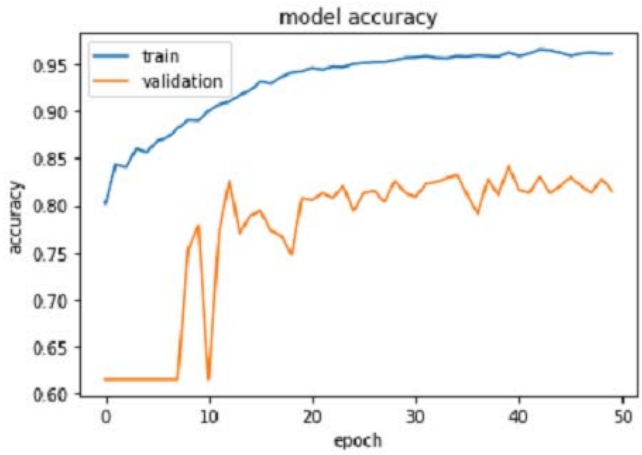

Fig. 5. Training and Validation Accuracy History

Fig. 6. depicts the history of training and validation loss. The model has comparative functionality in both the train and the validation dataset from the plot of loss. The training loss is decreasing exponential graph. It is almost constant after 30 epochs around 0.10 which is very less. Similarly, the validation loss is a spike having a maximum loss at the 4th epoch and the minimum loss is from epoch 10 to 16 and at epoch 44. The lower the loss, the stronger the model. The loss is measured on validation and training and here are two explanations of how it is given. The loss, unlike precision, is

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not a percentage. This is an addition to the errors prepared in each instance of the training or validation set.

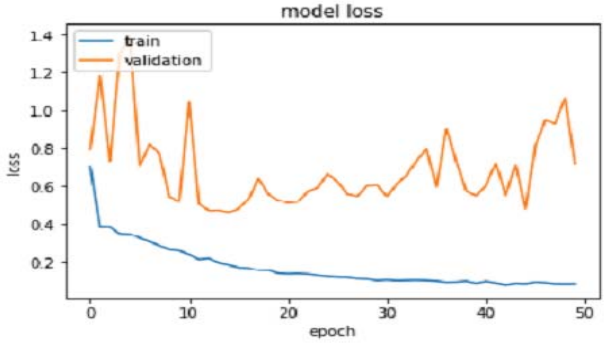

Fig. 6. Training and Validation Loss History

Receiver Operating Characteristic (ROC) as shown in Fig. 7. The ROC curve is most effective at the early stage of the assessment of the new diagnostic test. The ROC curve is drawn between True Positive Rate (TPR) and False Positive Rate (FPR). TPR means Sensitivity and FPR mean 1-Specificity. The AUC is an important way of summarizing the improved detection accuracy of the test [13]. It takes values from 0 to 1, where the value of 0 implies a perfectly incorrect test and the value of 1 represents a perfectly accurate test. The AUC can be measured using the trapezoidal rule 3. Overall, the AUC of 0.5 implies no prejudice, 0.7 to 0.8 is considered appropriate, 0.8 to 0.9 is considered exceptional, and higher than 0.9 is considered to be outstanding.

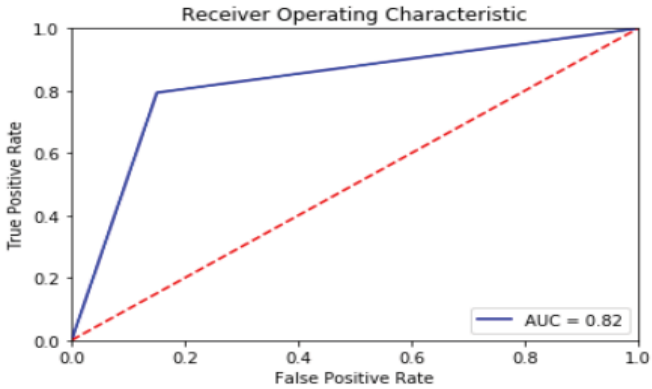

Fig. 7. Receiver Operating Characteristic

V. CONCLUSION

Most people suffer from diabetes, which is a main cause of DR. It is a dominant cause of blindness and, with proper treatment, its symptoms may be minimized. Therefore, a model must be developed that can detect DR without any expert guidance. Deep learning algorithm is a CNN that works effectively and accurately for the classification of images. In this experiment, we have compared the performance accuracy of an existing model. We have implemented the algorithm by modifying distinct parameters such as the number of convolution and pooling layer optimizer to achieve better efficiency. No manual feature removal steps are required in our model. Our network architecture has achieved adequate classification accuracy with dropout methods. True positive rate (or recall) is also improved. The future scope of this includes the model will be improved and trained on more datasets to be capable of identifying more diseases to help the physician in the clinic and hospitals.

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